

Republic Of Yemen
University Of Aden - Faculty Of Engineering
Odd Semester Examination - Academic Year 2022/2023

Class: B2IT& B2ITE

Date: 2/2/2023

Name: _____

Subject: Analog Electronics

Time: 2 hours

Reg. No.: _____

Answer question (1) (compulsory) and any three questions.

Question (1)

ترفق ورقة الأسئلة بملف الإجابة.

Choose the correct answer.

- a) A transistor can be operated as an electronic switch in
- Cutoff
 - Saturation
 - Both of them
- b) The ratio of dc collector current to dc emitter current in a BJT is
- α
 - β
 - Y
- c) Add a dc level to an ac voltage called
- Diode clampers
 - Diode clippers
 - Diode rectifier
- d) Rectifier is an electronic circuit that converts
- DC to AC
 - AC to DC
 - AC to AC
- e) In half wave rectifier, If the peak voltage is 20 V and load resistance is 2 K Ω ,
than the output DC current is
- 0.01A
 - 3.2 mA
 - 9.3 mA
- f) A center-tapped full wave rectifier connected to a transformer whose each
secondary coil has $V_{rms}=14$ V, $r_f=40\Omega$, $R_L=1k\Omega$ then the V_{dc} is
- 12.11V
 - 12.11mV
 - 19.79 V
- g) In cutoff, V_{CE} is
- 0 V
 - Equal to V_{CC}
 - Equal to V_c

h) The most heavily doped of the three semiconductor regions of a BJT is

- i. Collector ii. Base iii. Emitter

i) An example of the simplest form of diode clipper is

- i. Half-wave rectifier ii. Bridge rectifier iii. Center tap rectifier

j) The unit of efficiency is

- i. Watt ii. Volt iii. None of them

2# Put true or false and correct the mistakes.

- i. No matter what type of configuration a transistor is used, the base-to-emitter voltage is zero volt if the transistor is in the "on" state. ()
- ii. The percentage of DC component in rectifier output is called ripple factors ()
- iii. I_B is always greater than I_C ()
- iv. If I_C is 50 times larger than I_B than β is 100. ()
- v. If β times the value of R_E is at least 5 times the value of R_2 , the approximate approach can be applied with a high degree of accuracy. ()

Question (2)

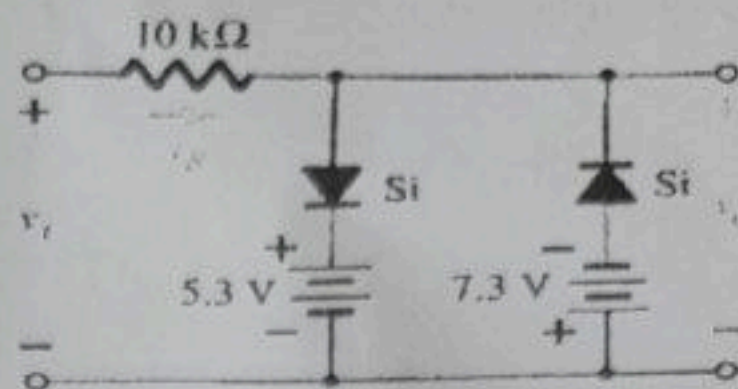
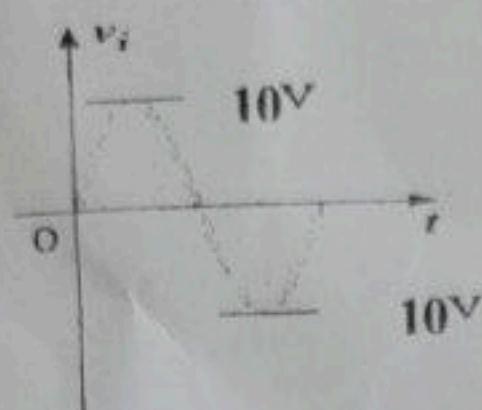
A center-tapped full wave rectifier connected to a transformer whose input r.m.s. voltage is 240 V. Assume the internal resistances of the diode and load resistance are 60 and 2k Ω , respectively and turn ratio 10:2.

a) Draw the center-tapped full wave rectifier circuit.

b) Determine: I_{dc} , V_{dc} , Rectification efficiency .

Question (3)

- Enumerate the DC biasing methods of BJT?
- What is the purpose of emitter resistor added to the DC bias?
- In figure shown below Determine the output voltage waveform.



Question (4)

- From clamping circuit with silicon diode in Fig.1 Determine the output voltage waveform.
- From Fig.2 what is the name of this circuit? Determine β , R_C , R_B , V_{CE} , V_E . (Where $I_C = 3.2 \text{ mA}$).

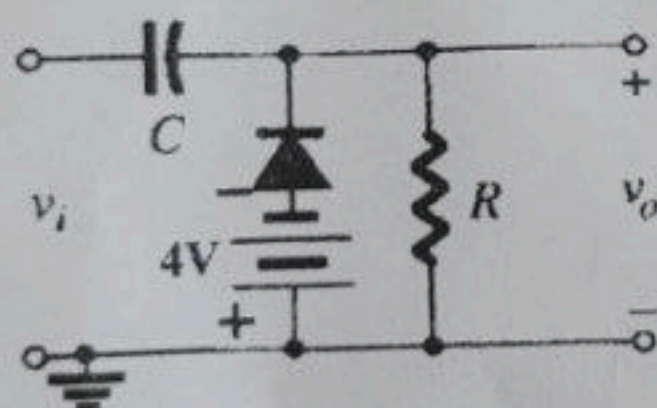
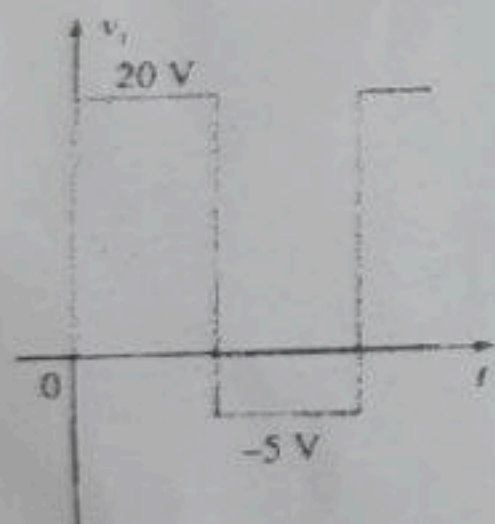


Fig.1

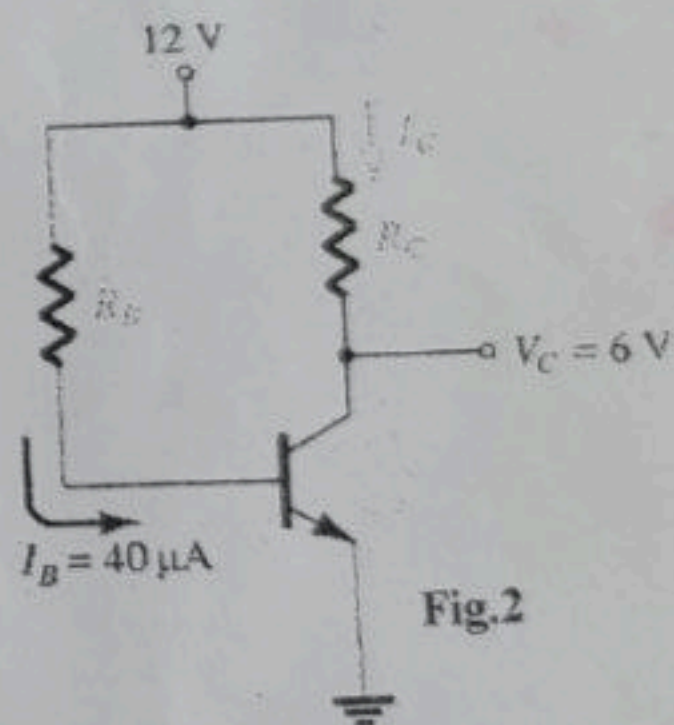


Fig.2

Question (5)

For the voltage-divider bias configuration of Fig.3

- Draw the Thevenin equivalent circuit
- Determine: I_B , I_C , V_{CE} , V_C .

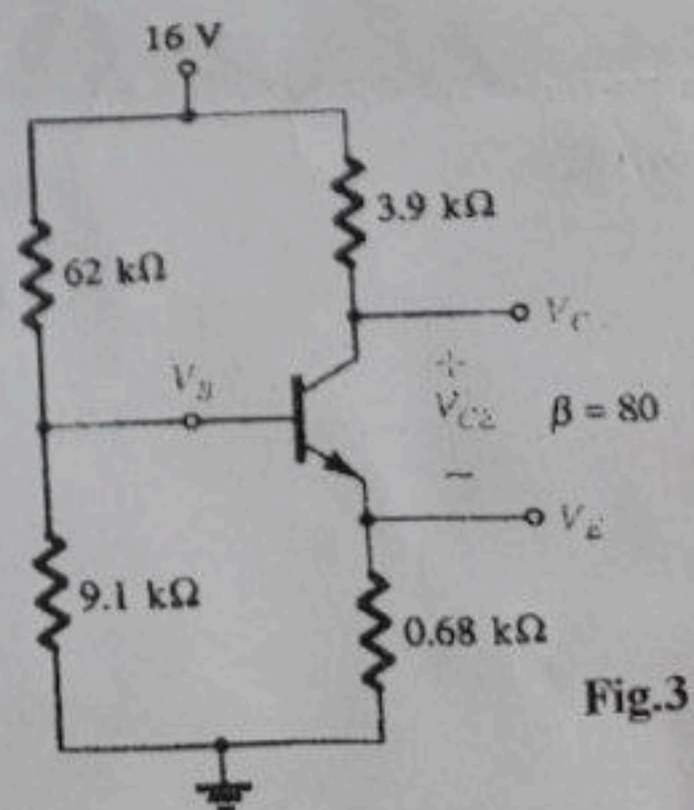


Fig.3